

Navy Exchange Website Usability Analysis

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Introduction

Throughout the duration of January and February 2023, I created a usability test kit and conducted a dozen usability tests for the Navy Exchange website (mynavyexchange.com). The tests sought to assess the ease of navigation and users' opinions of the site's content, along with examining any other pain points on the website. The 12 participants took the test in times ranging from 20 minutes to 50 minutes, many tests revealing repetitive behavior and providing valuable insight. This report aims to summarize these findings and offer recommendations to increase the overall user-friendliness of mynavyexchange.com.

Methodology Overview

The chosen methodology for the usability test was protocol analysis, a method where participants are asked to complete common tasks that users would expect to complete on a normal website visit. As participants navigated the website, they spoke out loud about what they were doing on the screen and any additional concerns or comments they had experienced. After completing these tasks, they were asked post-questions to evaluate their experience with the site.

After personally browsing the website, I identified important website information and tools and made note of potential navigation issues. The usability test kit was designed with the following site uses in mind:

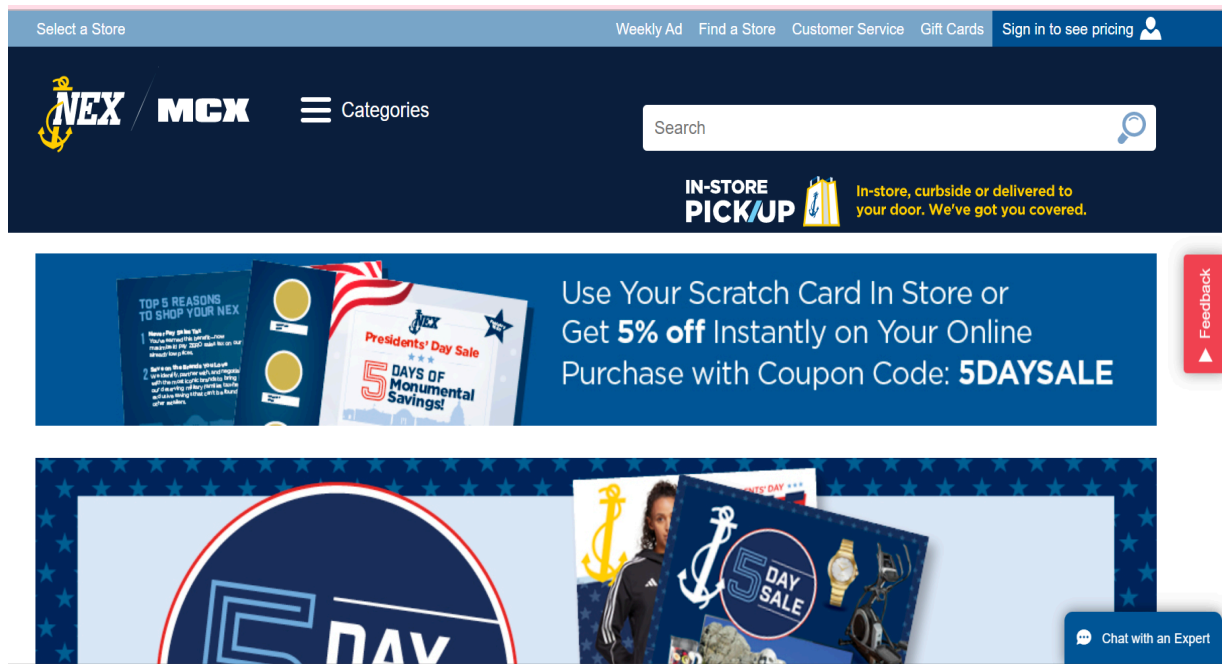
1. Find products and promotions on the site.
2. Access customer service information.
3. Utilize the tools provided by the site.

While checkout is one of the most crucial tasks on an e-commerce website, the checkout option was disabled for the sample account used for the usability test and thus could not be fully examined.

The chosen participants were meant to reflect the audience of the Navy Exchange. The participants ranged in age with five young adults in the 20 to 26-year-old age range, two adults in the 30 to 40 age range, and five older participants in the 45 to 65 age range. Age, however, did not play as big of a role in the ease of website usability as did the frequency that a participant shopped online. All the participants were military affiliated, half of them dependents and the other half active duty or retired service members. All 12 participants were given the same test, which can be found in the appendix. The sessions took place in a two-week period from January 24, 2023, to February 5, 2023, and all participants signed a consent form and were compensated with a \$50 NEX gift card. The findings and recommendations from these sessions are organized into three sections: the look and feel of the website, website navigation, and website content.

Website Look and Feel

Figure 1. Mynavyexchange.com homepage



Above is a screenshot of the Navy Exchange website homepage on a PC and an iPhone. All the participants unanimously agreed that the overall look and feel of the website were satisfactory. Apart from a few comments regarding the weekly ad (discussed in the following sections), there was overwhelmingly positive feedback. Some participants relayed their appreciation for the blue and gold color choice and the simple and uncrowded layout. Many participants said the website's look and feel were simply "fine", further describing it as standard and without issue. As such, there is no recommendation regarding the look and feel of the PC version of the website.

Website Navigation

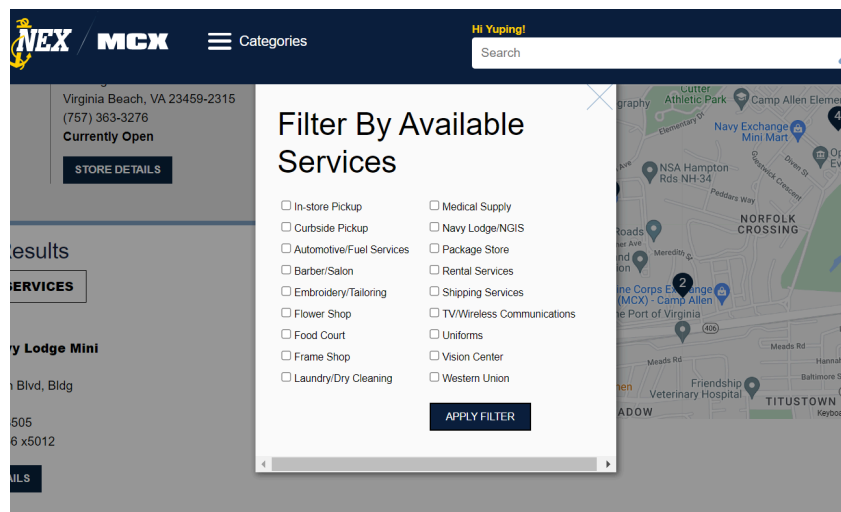
There are various navigation tools visitors could use on the NEX website, the main ones being the search bar, the pinned "categories" drop-down menu, and breadcrumb trails. These main navigational tools served to help most users, but the search bar proved to be occasionally inefficient. Additional tools such as the compare tool and the tool to find and select a store were also singled out for their complications. Below are recommendations to remediate the observed issues and improve the ease of navigation on the website.

Find a Store/Change a Store

After signing in to an account, the potential next step for many users is to find the stores closest to them. During the usability test, participants were instructed immediately after signing in to the sample account to select a store as “their store”. What was a seemingly simple task for some, became a complicated task for others. Complications include:

- While selecting a store as your own is an option in the upper left-hand corner, many users clicked on “Find a Store” instead of changing their current store. While somewhat emphasized by the bold yellow letters, it wasn’t always enough to draw users’ attention. This led those who needed to find the store closest to them to take a longer path.
- “Find a Store” does not prioritize listings unless prompted. The initial search by zip code, city/state, or country demonstrated no problems. However, once a user clicks “search”, every Navy Exchange affiliated store loads, with the closest stores coming first. Participants looking for a particular Main Navy Exchange store were given 24 locations within a 10-mile range, with the Main Navy Exchange store being the twelfth option. Some users stopped scrolling before they had even reached the store on the list. Users can filter the list by available services, which are shown below, but there is no specific nor obvious shopping-oriented keyword that would load the Navy Exchange. The one user who thought to use the filter tool quickly exited, stating that it was not very helpful for finding the Navy Exchange store they were looking for.

Figure 2. Filter By Available Services



- Users cannot see how far a store is without inputting the address. For users who chose to simply change their store, it lists the closest Main Navy Exchange stores in an order one has to assume is by distance. Apart from the assumption that the first result is the closest, there is no way of knowing how far the store is without

inputting the address. Given a range of 25 miles, the closest store could be 1 mile away or 24 miles away. The same problem persists under “Find a Store”, where the listings don’t provide an estimated distance from the input zip code, city, or country.

In response to these problems, the following suggestions are recommended to improve the search and selection of a store.

- Put greater emphasis on the store already selected or the “Change your Store” button. This could be done by moving it to a more highlighted part of the screen such as under the search bar. One could also move “Find a Store” directly next to “Change your Store”. Because these buttons are not next to each other, participants tended to click whichever one they happened to land on first. Moving these buttons directly next to each other would emphasize that both options are available on the pinned header without any additional search.
- “Find a Store” should prioritize the Main Navy Exchange stores in the search list. The primary function of the NEX website is to shop or browse products found at NEX and MCX stores. As such, when looking for a location on this website, it should be assumed that these main stores are the most sought. The closest Main Navy Exchange stores should be highlighted at the top of the search list and then the remaining store can be listed as is (by distance).
- The distance away from an input zip code should be added to each search listing. This will allow users to see an estimate of how close they are to the store.
- Similar to how the “Make My Store” button is available next to the “Details” button when a user is changing their store, the “Make My Store” button should be included in the location search results under find a store, at least for the Main Navy Exchange stores. Selecting a store is one of the more important functions for a user looking to purchase an item for pick-up or to browse the selections. Taking out the extra step to view all the store details makes the location search more efficient.

Search Bar

When instructed to find a certain product, most participants utilized the search bar as opposed to the “Category” navigation drop-down tool. Participants also attempted to use the search bar to find customer service information but were mostly unsuccessful. As the search bar is the first tool many users gravitate towards to navigate a website, it’s of utmost importance that it works efficiently. The search bar on the Navy Exchange website had only a couple of issues: the product-orientated search bar and the search result relevancy.

- The search bar is product-oriented. For the most part, it does not differentiate between searching for an answer on a webpage and searching for a product. Most participants searching for any customer service information were unsuccessful. However, using specific keywords would load a specific customer service page. Participants searching “Rewards” and “NEXt level” would only be met with random

products, but “rewards program” and “NEXt level rewards” would load the NEXt Level rewards customer service page. Searching for other customer service information in the search bar was also typically unsuccessful.

- The search bar does not always show the most relevant results first. Often, clearance items that are partially relevant appear at the top, and rarely, less relevant results appear at the top. There is no option to sort by relevancy. Despite the assumption that the original search would sort by the relevancy trait, if a user were to change the “sort by” trait and then wish to sort by relevancy, they would not find that to be an available option.

As such, the following suggestions are recommended:

- The search function should be able to differentiate product searches and simple information searches. Additionally, the search function should be able to better recognize inquiries that don’t necessarily fully encompass the search. For example, searching “rewards” should bring up the NEXt Level Rewards the same way that searching “rewards program” does. The same gray subheadings that appear in the drop-down menu while typing in the search bar (i.e. “categories” and “brand”) can be used to separate non-product information from product results.
- Add a “most relevant” option to the sort by drop-down menu.

Customer Service Page

Customer service information can be found on various pages on the NEX website, though most of it is concentrated under one customer service page. Navigating to this page is an issue on its own, with many of the participants skipping over the “Customer Service” link in the top bar, failing to find the information at the bottom of the screen, or unsuccessfully using the search bar. The formatting of the customer service page also led to frustration. To ease these frustrations the following are suggested:

- Add a search bar specifically to the customer service page. This page is particularly information-heavy and users should be able to find that information without reading through multiple sections.
- The subsections under “FAQs” on the left-side menu should be better differentiated from the main sections. This could be done by creating a drop-down menu or by further indenting the subheadings and changing the font size/color.
- “Shipping and Delivery Info” and “Policies & Programs” should be moved higher on the left-side navigation list.

Compare Tool

While the compare tool is a useful tool for seeing multiple products side by side, it likely isn’t used to its full potential. Almost all test participants struggled at least briefly when they were instructed to use the compare tool. One participant stated he had to use his

“outside-the-box thinking” to navigate this tool, the amount of effort a website should not require. Some of the complications with the compare tool are as follows:

- Many participants clicked initially on the “compare selected” button without selecting any items expecting a page or any type of pop-up that would allow them to use the tool. The position of the compare selected button at the top of the page would indicate to users that it should be selected first when that is not the case.
- Initially, finding the “add to compare” checkbox was difficult for participants. It blends in with the description of the item and was overall very easy to overlook. Many participants needed redirection, as they searched the entire page or went to different pages before noticing the checkbox.
- The “add to compare” button is not available on the actual product page. If a user wants to learn more about a product, they’ll have to navigate back to the previous page afterward to add it to the compare tool, an unnecessary step.
- After a user has selected a maximum of four products to compare, it does not allow any more boxes to be checked. The notice for the maximum, however, only appears at the top of the page under the “compare selected button”. Essentially, if the next product being selected to compare does not appear in the first or second row, this notice would not be seen by users.
- There was a repeated issue where the “compare selected” button would stop working after a user navigated a couple of pages. Despite the correct number of boxes being checked, a blank white screen would appear and the compare tool would not allow for comparison. Users had to unselect and recheck the “add to compare” boxes before the tool would properly function. Similarly, when a user removed a product from the compare tool, the same blank screen would appear.

To create a more productive and accessible comparison tool, the following is recommended:

- If the compare selected stays in the position at the top of the screen, it should have a pop-up notice that the boxes need to be selected in order to compare. Similar to how the notice of the maximum selected product shows up under the button now.
- The “add to compare” should be emphasized on the product list page. This could be done by enlarging the “add to compare” text and checkbox and bringing it to the top of the description under the photo. Another alternative is to replace the checkbox with an “add to compare” button that looks similar to the blue “compare selected” button. Then, change the text to “added to compare” and the button color to yellow when it has been selected.
- Make the “add to compare” button available on the information page for each product.
- The notice for the maximum products selected for the compare tool should appear directly next to “add to compare” when a user attempts to add a fifth product.

- The error that leads to a blank screen should be patched.

Miscellaneous Navigation Issues and Suggestions

- One of the most frustrating issues for participants was looking for a product and then finding that it wasn't available for pickup or shipping to the selected store. While there is an option to narrow the products shown by what is available for pick up the next day, there is no option to narrow down what is available to ship to a user's home. Furthermore, the option to narrow by "pickup tomorrow" was often missed. Many participants expressed that if an item is not available for pickup or shipping, it should either not be shown in the search results or it should be marked unavailable on the products page before a user clicks the product to read its information.
- There were some occurrences where clicking on "weekly ad" would open the weekly ad in another tab and sometimes on the same page. Participants who selected the weekly ad where it had opened on the same page had more difficulty returning to the homepage compared to participants where the weekly ad opened in a new tab. In both cases, however, navigation back to the NEX homepage was a slight issue. The shift of the NEX logo position from the top left on the Navy Exchange homepage to the top right on the weekly ad homepage confused participants who were acclimated to clicking in the top left to return to the page.
- On the NEXt Level Reward page, selecting "create an account" takes users back to the homepage if they are already logged in to an account. This led participants to believe the button didn't work, forgetting they were already signed in. It is suggested to link this button to the account settings of users already signed in, instead of rerouting them back to the homepage.
- While signing in was a typically quick and easy task, some participants commented on the inconspicuous appearance of the button. While it is a different color, it is not highlighted well. The Navy Exchange should consider changing the fill color to a more contrasting and bold color, such as the NEX gold found on the rest of the page.
- Under the "Riding Toys & Outdoor Play" category, there are two "skateboards" subcategories. They both include the same products, the only difference is that one is titled "SKATEBOARDS" in all caps.
- One of the categories under household essentials is titled "Rodant Control" when it should be titled "Rodent Control".
- While users can search for brands in the search bar, some users attempted to select the brand name on the product page anticipating it would direct them to a page with that brand's products. NEX should consider linking the brand name on the

product pages to the same brand page that loads when a user searches for the brand name in the search bar.

Website Content

While NEX offers a wide range of services, the majority of users who visit the NEX website are likely searching for products available at the main Navy Exchange stores. Test participants believed that the product selection was sufficient and while there may have been issues with navigating to specific content, all necessary information was present on the website. What most participants did experience trouble with, however, was the formatting of a few pages and the value and understanding of the NEX Level Rewards program.

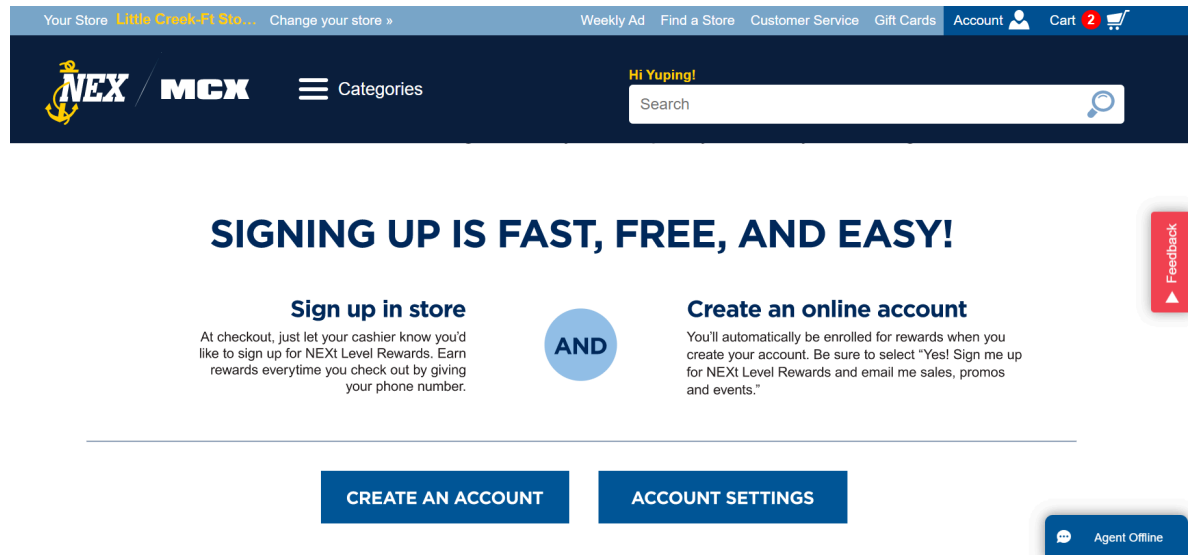
User-Friendly Formatting

There were two sections where participants voiced their dissatisfaction with the readability and formatting of the page: the Weekly Ad and the Customer Service page.

- The Weekly Ad in its page-view format is not very website-compatible. It seems to be an uploaded PDF version of the physical weekly ad materials. Participants pointed out that the text was extremely small to read and the Weekly Ad page did not feel cohesive with the rest of the website. While the full-screen option or “contents view” would have made the Weekly Ad a bit easier to read, not a single participant used, or perhaps even noticed, these options. Furthermore, it doesn’t solve the issue of the incohesive feel and formatting that essentially redirects users to a new website. The Navy Exchange should consider changing the Weekly Ad page to a page similar to the home screen. It should follow the same flow and formatting as the other pages on the NEX website. Likewise, users should be able to stay on the same NEX website when viewing the weekly ad instead of being redirected to a digital publication. Users can still have the option to download the PDF of the weekly ad, but it should not be the default view.
- The readability of the customer service section is wordy, however, this may be unavoidable due to the amount of information that needs to be present. The policies and programs page was singled out as an example of users having too much to read to find the important information. The long and unbroken sections should attempt to break up the information and use bullet points when needed. More important information/frequently searched information should be bolded or highlighted. For example, under the return section, the **45-day return policy** should be emphasized.

NEXt Level Rewards Program

Figure 3. NEXt Level Rewards Sign-up directions



- One of the biggest issues when participants viewed the NEXt Level Rewards Program was understanding the two main steps to sign up for the program, pictured below. Despite the bold “AND” that implied both steps need to be done, ten out of twelve of the participants read “OR”. The eleventh participant initially read “AND” and then proceeded to read “OR”. This pre-assumption is perhaps because signing up in-store *and* online is not the usual procedure for joining a rewards program. Nonetheless, this assumption seems to overshadow the actual instructions. This should be changed to a step one and step two format instead of the “AND”. It should also be formatted vertically where step one is above step two and the steps are bolded. This would better emphasize that it is a process that requires one action after another.
- The rewards collected by NEXt Level members are not easily trackable. There is no way to view rewards under a user’s account as they only receive the rewards through email. The NEX should consider creating a section under “My Account” dedicated to the rewards program. It should include upcoming available rewards, points tracking, and opportunities to earn more rewards/points to minimize the time it takes to receive rewards, maximize return for users, and as such, encourage more engagement and spending from users.
- The NEXt Level Rewards were simply not worth it to many participants. The rewards were low value, with many pointing out that the main perk of joining had such a small return (2%) for spending a large amount. While less relevant to the content offered, this repeated verbal dissatisfaction with the value of the rewards program was worth mentioning.

Conclusion

The usability test conducted for the Navy Exchange website uncovered various pain points that may have hindered users' experience. Website navigation specifically will need attention from the NEX staff. While some of these issues may be easy to fix, such as typos or broken links, others may require a bit more effort to resolve. Nonetheless, the overall user experience for the website was unanimously positive, with all participants being able to eventually find the needed information with a few bumps in the road. Given the findings and suggestions of this report, the Navy Exchange can take its first steps in providing a smoother, more efficient user experience.

Appendix I. Usability Analysis Test Kit

Test Kit for All Users:

I am going to read the instructions to you.

Introduction (To be read to each test taker)

Thank you for coming today. We need your help in assessing the content and the user-friendliness of the Navy Exchange's website.

You will be given eight tasks to complete while browsing the website. As you browse the website and perform the tasks, please describe verbally what you are doing and what you are thinking. For example, if you are looking for a link on a page, please tell us what you are looking for and why you are looking for it. As another example, if you click on a link, please say which link you have just followed and what you are expecting.

As you are browsing the website, we might remind you to think out loud. If you have comments about the look and the organization of the website, feel free to say them too. And if you have any other comments that are not covered by the task list, please mention them as well.

The computer screen will be videotaped and your voice will be audio-taped. Please speak out loud so that I can hear you. The program will take one hour to complete. Your information will be kept confidential and is purely meant for research purposes only. Please ask us to stop whenever you feel tired. You can quit at any time if you feel uncomfortable. Please understand that we will evaluate the website's quality, not how well you browse the site. So relax and be honest. Your responses are highly valuable to us.

Please sign below if you agree to participate in the research.

Signature: _____ Date: _____

Before we begin the main tasks, I'm going to ask just a couple of pre-questions.

Pre-Questions:

- 1) What is your affiliation with the military?
- 2) How often do you shop on the internet?
- 3) How frequently do you shop at the Navy Exchange?
- 4) Have you ever shopped at the mynavyexchange.com website? What have you bought there?

To familiarize you with the process, we have some initial tasks we would like you to do. We will now start the initial tasks. Please remember to talk out loud.

- 1) Go to the Navy Exchange website (<https://www.mynavyexchange.com/>)
- 2) On the homepage, click on "Categories" and view the categories of goods that the Navy Exchange website offers.

Now that you are familiar with the procedure, let's continue with the eight tasks. Please go through the task listed below one by one starting with the first task. When you are done with one task, verbally confirm you have finished the task, and then go on to the next task. I will also be reading the questions aloud. Please remember to talk out loud.

- 1) On the NEX website, log in to an account. Sign in with the following information:
 - email: yxxliu@odu.edu
 - password: ODUrocks@2021

- 2) Suppose you are in the 23508 zip code and would like to change your store on the NEX website. Please use the NEX website to make that change. Select the **Main Navy Exchange** store that is the closest to the 23508 zip code as your main store. Write the name of the store below.

- 3) Using the NEX website, view the weekly ad. Find two items in the weekly ad and write down the regular price and the on-sale price.
 1. Item: _____
 Regular Price: _____ Sale Price: _____

 2. Item: _____
 Regular Price: _____ Sale Price: _____

- 4) Imagine you are shopping for a Nintendo Switch, which is a handheld videogame console. On the NEX website, find a Nintendo Switch under \$200. List the price of the product, its two features, and whether it is available for curbside pickup at the main store you selected earlier

Price: _____

Feature 1: _____

Feature 2: _____

Available for curbside pickup? _____

- 5) Using the NEX website, compare **side by side with the compare tool** the 3 following products:
 - "NINTENDO SWITCH OLED"
 - "NINTENDO SWITCH WITH NEON BLUE NEON RED JOY-CON"
 - "NINTENDO SWITCH LITE - BLUE"
 Looking at the side-by-side comparison, write below which product (of the three) is the most expensive and one unique feature that it has compared to the others.

- 6) Find the cost for standard shipping per order and how long it may take for shipping to the Continental United States.

Cost: _____

Shipping Time Estimate: _____

- 7) Find the general return policy for the NEX website and write below how many days you have to return an item purchased from the website.

_____ days

- 8) A customer decides to join the rewards program. Figure out how to sign up for the NEXt Level Rewards and write it down below.

The customer wants to put the program to use. Write below how many points you would receive for each dollar of purchase and what you receive for your points.

Post-Questions:

- 1) Were you able to find all the information you needed?
- 2) Was it easy to find the information? If not, why?
- 3) What do you think about the company's website?
- 4) How do you evaluate the look and feel of the company's website?
- 5) What would you suggest to the company to improve the website?
- 6) Would you use this website or recommend it to your friends in the future?